

Deep Learning Framework for Medicinal Leaf Classification Using Transfer Learning

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Abstract—Medicinal plants are integral to traditional healthcare practices like Ayurveda, offering therapeutic value at minimal side effects. The identification of medicinal plant species, however, presents the biggest challenge as the species exhibit morphological similarity, few botanists, and the risk of misplacement. In response, the current study puts forth an extensive deep learning-based transfer learning framework for the automatic classification of medicinal leaves. The BDMediLeaves dataset, made available for the public, consisting of 38,606 augmented images of ten species, was adopted in the development and validation of the framework. Five pre-trained CNN models—VGG16, DenseNet201, InceptionResNetV2, MobileNetV2, and Xception—were experimented in an overall experiment setup to evaluate their performance based on accuracy, precision, recall, and F1-score as the basis of their performance. The MobileNetV2-based proposed model delivered the maximum accuracy of 99.43%, performing better than the other models while ensuring computational efficacy. The study's novelty lies in establishing a systematic and deployment-ready benchmarking framework for medicinal plant identification, demonstrating that lightweight CNNs can achieve state-of-the-art accuracy suitable for mobile and web-based healthcare applications.

Keywords— Medicinal plants, Leaf Classification, Transfer Learning, Deep learning, Image classification, Convolutional neural networks

I. Introduction

In the current world, increasing disconnection from nature impacts the health and well-being of humans. Traditional systems, such as Ayurveda, which center around medicinal plants, provide sustainable, side-effectless treatment alternatives that tackle diseases at their outset [1]. Their uses extend to pest control, natural coloring, food products, teas, and fragrances. The world, at over 50,000 species, relies on plants for health-associated products, where India and China take the lead at 7,500 and 11,146 species, respectively, and other nations exploit 7–44% of their flora for medicines [2]. This world dependence gives rise to the imperative of correct identification due to the health risk and sustainability hampered by counterfeit and poor-quality herbal formulations.

Leaves, with readily extractable features like shape, color, and texture, offer a practical foundation of classification [3]. However, conventional expert-based identification is time-consuming, subjective, and in appropriate form for broad-based applications. Developments in computer vision and image processing now enable fast, automatic, and very accurate classification, bringing plant identification to the non-expert's desk.

To overcome limitations in traditional methods, the current study devises an end-to-end deep learning framework that can be deployed for automatic medicinal leaf classification. The proposed method utilizes transfer learning to extract discriminative visual patterns in an efficient yet computational light way that can accommodate use in the web and smartphone domains. Five pre-learned convolutional neural network (CNN) models, VGG16, DenseNet201, InceptionResNetV2, MobileNetV2, and Xception, is fine-tuned and rigorously tested using accuracy, precision, recall, and F1-score over the BDMediLeaves database. Differing from previous research that solely relied upon accuracy, the current study presents a standardized methodology for benchmarking the architectures by comparing the architectures under the same experiment settings so that their trade-offs in accuracy against efficiency can be ascertained.

The main contributions of this work are:

1. A reproducible and systematic assessment structure of five transfer learning models for the classification of medicinal leaves based on the big-data BDMediLeaves collection (38,606 images).
2. A benchmark study presenting model quality under a common environment, with fairness and equalability maintained in all architectures.
3. Obtaining the overall maximum accuracy of 99.43% by MobileNetV2, confirming its applicability for light, high-accuracy medicinal plant recognition.
4. Development of a deployment-focused framework that can accommodate web and mobile-based applications, thus connecting the gap between practice and research.

The remaining paper is structured as follows. Section II presents the literature review, Section III details the proposed framework and methodology, Section IV discusses results and analysis, and Section V concludes the study with potential future work.

II. Literature Review

Research on medicinal plant classification has evolved significantly over the past decade, progressing from traditional image processing techniques to advanced deep learning approaches. Early studies relied on handcrafted features—such as color, texture, and shape—combined with machine learning classifiers like Support Vector Machines (SVMs) for plant identification. While effective for small datasets, these methods often struggled with scalability and accuracy on large, diverse image collections.

A. Traditional Machine Learning Approaches

The early classification of medicinal plants was based on hand-based features (color, texture, form) with the classifiers such as SVM and ANN. Kan *et al.* [4] got 93.3% accuracy for 12 species based on the features of the shape and texture. Habiba *et al.* [5] got 96.11% based on the texture features of Modified Local Gradient Pattern-based in 1,054 images. Pacifico *et al.* [6] got 97% based on color and texture features extracted from 1,148 images, while Azadnia *et al.* [7] got 100% based on six species with the help of ANN. Other SVM-based research got 90% [8] and 91% [9] accuracy.

B. Transition to Deep Learning

CNNs amplified scalability and accuracy by automating feature extraction. MobileNet was employed by Trung *et al.* [10] to classify ten classes (98.7%), Valdez *et al.* [11] used MobileNetV3 for classification of 11 classes (97.43%), and Raisa *et al.* [12] classified 37,693 images of Bangladeshi medicinal plants (71.3%). The accuracy was 99% by ResNet50 [13], and Saikia *et al.* [14] used a neural network to classify six species based on 90 RGB images.

C. BDMediLeaves Dataset Studies

Islam *et al.* [15] applied DenseNet201 and InceptionResNetV2, achieving 80.69% and 90.09% accuracy. Himel *et al.* [16] proposed MedPlantNet with spatial attention and multi-scale fusion (98.51%), Shanto *et al.* [17] achieved 94.11% with CNN-SVM, and Islam *et al.* [18] developed MSFFAR-Net with MobileNetV2 and an attention module (98.12%).

D. Ensemble and Custom Architectures

Uddin *et al.* [19] tested multiple CNNs (VGG16, ResNet50, DenseNet201, InceptionV3, Xception) and three custom networks, with ensembles reaching up to 99% accuracy. Bharadwaj *et al.* [20] introduced Felicia based on Inception-V3 (99.35%), Dileep *et al.* [21] combined AlexNet with Softmax/SVM (96.76%), and Kiflie *et al.* [22] used transfer learning for Ethiopian species (VGG19, 94%).

E. Explainable and Hybrid Models

M.T. Islam *et al.* [23] proposed a ResNet50-PSO-SVM cascade for seven species (99.60%). Widneh *et al.* [24] trained MobileNet on 15,100 images from 52 species (92%). Sharma *et al.* [25] evaluated VGG16 and VGG19 for plant disease classification, with VGG16 achieving 98.52%.

Conventional ML approaches are good enough for small datasets, while deep learning, particularly transfer learning, achieves higher accuracy for big-data medicinal plant classification. Compact CNNs such as MobileNet reach nearly state-of-the-art accuracy, and attention or ensemble methodologies can also increase performance. The present study expands these developments by evaluating five pre-trained CNN models against the BDMediLeaves dataset.

III. Proposed Framework and Methodology

This section presents the intended deep learning framework and experiment approach designed for the automatic classification of leaves of medicinal plants. It elucidates the characteristics of the adopted dataset, workflow of the

framework, transferred learning architectures adopted, and the metrics used in validating the performance of the model.

A. Dataset Description

The experiments were performed over the available BDMediLeaves dataset [15], consisting of leaves of ten species of medicinal plants that are popular in Ayurvedic and traditional medicines. The initial set of 2,029 images was augmented by means of data augmentation, i.e., random rotation, flipping, zooming, and shifting, to create 38,606 balanced samples, thus increasing the generalization and the minimization of the class imbalance. The images were of the same size, i.e., 224×224 pixels, and all the images were normalized in the [0, 1] range to obtain uniform input representation by all the models. The set was divided into 70% training, 15% validation, and 15% test subsets.

B. Proposed Framework

The suggested deep learning architecture for the classification of medicinal leaves includes five major steps, data preprocessing, augmentation, feature extraction, fine-tuning, and evaluation, guaranteeing systematic learning and reproducible outcome.

i. Data Preprocessing and Augmentation

The BDMediLeaves was resized first to 224×224 pixels to retain compatibility with pre-trained CNN models. The pixel intensity was scaled in the range 0 to 1 by doing normalization of the images. Data augmentation in the form of random rotation, horizontal and vertical flipping, zooming, and shifting was carried out to increase the dataset from 2,029 to 38,606 images. This maintained class balancing and enhanced the generality of the model under varying illumination and the orientation conditions.

ii. Feature Extraction using Transfer Learning

Pretrained models (VGG16, DenseNet201, InceptionResNetV2, MobileNetV2, and Xception) were initialized by ImageNet weights. Early convolutional layers were frozen to retain common low-level features such as textures and edges, while the late layers were fine-tuned to learn species-special properties of medicinal leaves.

iii. Fine-Tuning and Classification

Global Average Pooling, Dropout (0.5), and a Dense layer with softmax activation was added as a fully connected layer for 10-class classification. The models were optimized by Adam optimizer with learning rate $1e-4$, categorical cross-entropy loss, and early stopping with patience of 15 epochs to prevent overfitting. The training was carried out for maximum 50 epochs with a batch size of 32.

iv. Evaluation and Comparison

Performance was evaluated over the test splitter (15%) by means of accuracy, precision, recall, F1-score, and confusion matrix. The outcomes were checked over the models to ascertain the best architecture both in accuracy and in computational cost.

v. Deployment Potential

The last model, MobileNetV2, showed better performance (99.43%) with low parameter numbers and inference time,

yet was ready for deployment on the web and in mobile phones. This supports the usability of the suggested framework in practical-use identification of real-world medicinal plants.

C. Transfer-Learning Architectures

Five pre-trained CNNs, i.e., VGG16, DenseNet201, InceptionResNetV2, MobileNetV2, and Xception, were utilized for fine-tune learning over the BDMediLeaves dataset.

- VGG16 uses uniform 3×3 convolutional blocks suitable for hierarchical feature extraction.
 - DenseNet201 supports dense connectivity, encouraging feature reuse and gradient propagation.
 - ResNetV2's Inception combines multi-scale feature extraction along with residual shortcuts for training efficiency.
 - MobileNetV2 uses the inverted residual structure with bottleNecks that are linear, providing a light architecture suitable for mobilized deployment.
 - Xception applies depthwise separable convolutions, making it more efficient without sacrificing accuracy.
- These architectures represent a diverse balance of depth, computational complexity, and efficiency, allowing comprehensive evaluation of model adaptability to medicinal leaf images.

D. Evaluation Metrics

Performance was evaluated by standard measures: accuracy, precision, recall, and F1-score, calculated for the test split. A confusion matrix was produced for viewing per-class prediction accuracy and misclassifications. These quantitative and qualitative analyses provide a robust understanding of each model's reliability in distinguishing morphologically similar species.

IV. Results and Discussion

This section presents an in-depth performance evaluation of the experimented transfer learning models to classify medicinal leaves. The models were all set initially to train for 50 iterations, and early stopping was set to avoid overfitting by monitoring loss at training and validation, with the patience set at 15. The models were trained at a batch size of 32. The models' performance was evaluated based on accuracy, precision, recall, F1-score and confusion matrix. The result comparison of the five models under consideration has been explained in the following subsections.

A. VGG16

The VGG16 network performs robust and stable all-around performance for all ten classes of medicinal plants, with recall and precision values mostly at or around 1.00 (Table I). Low points are seen in recall for Hibiscus rosa-sinensis (0.98) and in recall for Kalanchoe pinnata (0.98), and in precision for Centella asiatica (0.98) and in precision for Ocimum tenuiflorum (0.98). However, F1-scores are high (≤ 0.99 in most of the cases), reflecting few false positives and false negatives. Generally, the findings establish VGG16's superior classification ability and strong generalization ability across varying morphologies of leaves.

TABLE I. CLASSIFICATION REPORT - VGG16

	Precision	Recall	F1-Score	Support
Azadirachta indica	1.00	1.00	1.00	595
Calotropis gigantea	1.00	1.00	1.00	580
Centella asiatica	0.98	1.00	0.99	589
Hibiscus rosa-sinensis	1.00	0.98	0.99	554
Justicia adhatoda	0.99	1.00	1.00	590
Kalanchoe pinnata	1.00	0.98	0.99	586
Mikania micrantha	0.99	1.00	0.99	613
Ocimum tenuiflorum	0.98	0.99	0.99	591
Phyllanthus emblica	1.00	1.00	1.00	545
Terminalia arjuna	1.00	0.99	1.00	581

Fig. 1 confusion matrix indicates the VGG16 model performed steadily high, with almost all the sample correctly classified among the ten species of medicinal plants. Most of the prediction falls onto the diagonal, reflecting good accuracy and well-balanced class performance. The misclassifications are few, occurring in a handful of sample in classes such as Hibiscus rosa-sinensis and Kalanchoe pinnata, and they do not really disrupt the overall result. The same is in agreement with the classification report, verifying the precision, recall, and F1-score around 1.00 for all the classes.

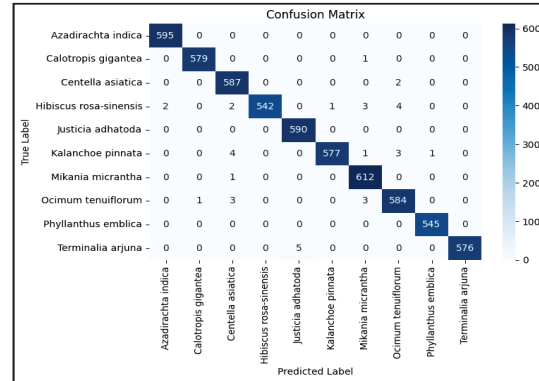


Fig. 1 Confusion matrix for VGG16 model

B. DenseNet201

The DenseNet201 classification summary (Table II) shows robust performance in general, as many of the classes got precision, recall, and F1-measures above 0.95. Phyllanthus emblica nearly performed flawlessly (F1-measure ≈ 0.997), whereas Azadirachta indica and Calotropis gigantea exhibited good trade-off among precision and recall as well. Low values were observed for Centella asiatica (recall ≈ 0.88) and Ocimum tenuiflorum (precision ≈ 0.90), indicating periodic mistakes. Although these fluctuations presented, the model retained high accuracy in all the classes, thereby validating the applicability of DenseNet201 in the classification of medicinal plant leaves.

TABLE II. CLASSIFICATION REPORT - DENSENET201

	Precision	Recall	F1-Score	Support
Azadirachta indica	0.98	0.98	0.98	591
Calotropis gigantea	0.97	0.99	0.98	585
Centella asiatica	0.96	0.88	0.92	583

Hibiscus rosa-sinensis	0.99	0.95	0.97	552
Justicia adhatoda	0.95	0.98	0.96	584
Kalanchoe pinnata	0.93	0.98	0.95	588
Mikania micrantha	0.99	0.91	0.95	625
Ocimum tenuiflorum	0.90	0.97	0.93	588
Phyllanthus emblica	0.99	0.99	0.99	544
Terminalia arjuna	0.96	0.96	0.96	584

Fig. 2 presents the confusion matrix for DenseNet201, indicating overall robust classification accuracy with high numbers of correct predictions for the majority of the classes. However, some of the classes—like *Centella asiatica*, *Ocimum tenuiflorum*, and *Mikania micrantha*—present distinct confusions, and errors are spread over several other classes. Even though the precision and the recall values are high overall, the confusions imply that the species with similar appearance attributes or texture of background presented higher challenges for the model in relation to others.

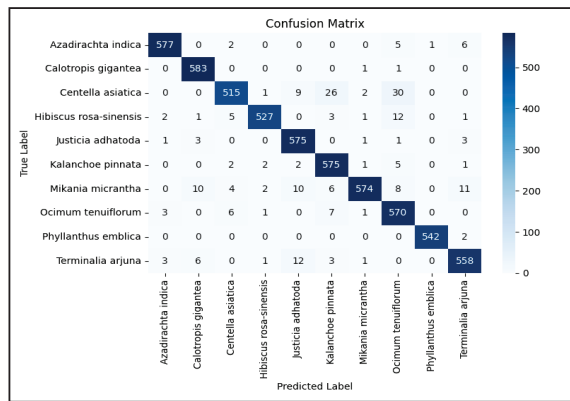


Fig. 2 Confusion matrix for DenseNet201 model

C. InceptionResNetV2

The InceptionResNetV2 classification report (Table III) shows close-uniform and very high performance in all ten classes of medicinal plants, with accuracy, recall, and F1-measures all in the range of 0.99 or higher. Small differences, like *Mikania micrantha* having slightly reduced precision (0.98), are insignificant and do not really affect the overall model performance. This uniform accuracy in very diverse classes shows that InceptionResNetV2 performed very well in selecting discriminative features, keeping both sensitivity and specificity, and reducing misclassifications.

TABLE III. CLASSIFICATION REPORT -INCEPTIONRESNETV2

	Precision	Recall	F1-Score	Support
Azadirachta indica	0.99	1.00	0.99	591
Calotropis gigantea	0.99	0.99	0.99	588
Centella asiatica	0.99	0.99	0.99	584
Hibiscus rosa-sinensis	0.99	0.99	0.99	568
Justicia adhatoda	0.99	0.99	0.99	587
Kalanchoe pinnata	0.99	0.99	0.99	586
Mikania micrantha	0.98	0.99	0.99	613
Ocimum tenuiflorum	0.99	0.99	0.99	586
Phyllanthus emblica	1.00	0.99	0.99	540
Terminalia arjuna	0.99	0.99	0.99	581

The InceptionResNetV2 classification report, reinforced by the confusion matrix in Fig. 3, shows close to perfect performance for all plant species. Most classes' precision, recall, and F1-scores stay steadily at 0.99, while the *Phyllanthus emblica* gets a spotless precision of 1.00. The confusion matrix presents low numbers of misclassifications, where only a single to three samples at a class end up being mislabeled in few cases. These findings show that the fine-tuned InceptionResNetV2 reliably generalizes to novel data while keeping stable performance in all these classes, an indication of good feature extraction and separation among classes ability.

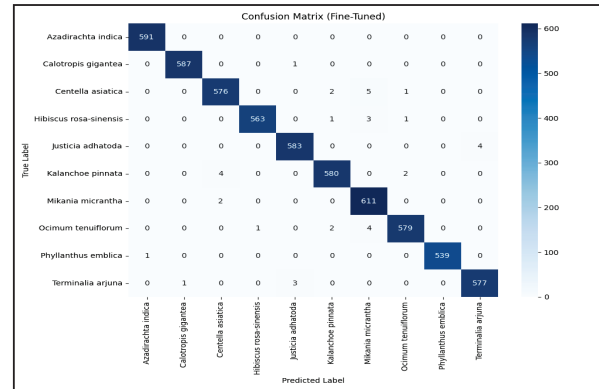


Fig. 3 Confusion matrix forInceptionResNetV2 model

D. MobileNetV2

The MobileNetV2 classification report presented in Table IV indicates outstanding performance in all the classes, with most of the precision, recall, and F1-values above 0.99. *Phyllanthus emblica* recorded a total score of 1.00 in all the metrics, while species *Azadirachta indica*, *Calotropis gigantea*, and *Terminalia arjuna* also recorded nearly-perfect scores with small deviations. Somewhat lower yet still very high scores were registered in the case of *Centella asiatica* and *Hibiscus rosa-sinensis*, advising a few wrong-classifications yet no substantial performance reductions. Generally, MobileNetV2 presents strong generalization, upkeeping balanced accuracy in all the species of plants with strong computational efficiency.

TABLE IV. CLASSIFICATION REPORT - MOBILENETV2

	Precision	Recall	F1-Score	Support
Azadirachta indica	0.99	1.00	0.99	594
Calotropis gigantea	0.99	1.00	0.99	570
Centella asiatica	0.98	0.99	0.99	583
Hibiscus rosa-sinensis	0.99	0.99	0.99	580
Justicia adhatoda	0.99	0.99	0.99	580
Kalanchoe pinnata	1.00	0.99	0.99	576
Mikania micrantha	0.99	0.99	0.99	588
Ocimum tenuiflorum	0.99	0.99	0.99	569
Phyllanthus emblica	1.00	1.00	1.00	580
Terminalia arjuna	1.00	0.99	0.99	577

The MobileNetV2 confusion matrix (Fig. 4) indicates that the model performed nearly perfect classification performance for all ten medicinal plant classes. The majority of the classes contain zero or close-to-zero misclassifications, with very

few cases (1–5) being misclassified as belonging to other classes. The diagonal dominance indicates the high recall and precision, and the few mistakes made are dispersed without a systematic confusion pattern, which shows the model can distinguish among visually similar species well. This is consistent with the high recall, high precision, and high F1-scores in the classification metrics, which verify the robust and generalization ability of MobileNetV2 for the current dataset.

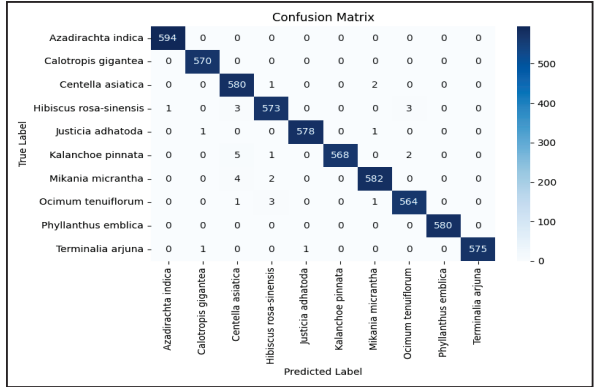


Fig. 4 Confusion matrix for MobileNetV2 model

E. Xception

The classification report of the Xception model (Table V) shows steadily very high precision, recall, and F1-scores for all the medicinal plant classes, the majority being higher than 0.98. The recall and precision were perfect for the case of Phyllanthus emblica, while other classes like Justicia adhatoda, Calotropis gigantea, and Mikania micrantha also exhibited nearly perfect performance. Small declines in recall for the cases of Hibiscus rosa-sinensis and Kalanchoe pinnata indicate some misclassifications, yet these were insignificant in affecting the overall good performance. In general, the findings show the capability of Xception in coping with inter-class visual resemblance while having fair-balanced accuracy and generalization.

TABLE V. CLASSIFICATION REPORT - XCEPTION

	Precision	Recall	F1-Score	Support
Azadirachta indica	0.99	0.99	0.99	591
Calotropis gigantea	0.99	0.99	0.99	587
Centella asiatica	0.98	0.99	0.99	594
Hibiscus rosa-sinensis	0.99	0.97	0.98	558
Justicia adhatoda	1.00	0.99	0.99	582
Kalanchoe pinnata	0.99	0.98	0.99	578
Mikania micrantha	0.99	0.99	0.99	617
Ocimum tenuiflorum	0.97	0.99	0.98	587
Phyllanthus emblica	1.00	1.00	1.00	545
Terminalia arjuna	0.99	0.99	0.99	585

Fig. 5 demonstrates that the Xception model performed nearly perfect classification across all ten species of medicinal plants, with the majority of the predictions along the diagonal with minimal misclassifications. Small mistakes, like frequent misidentification of Hibiscus rosa-sinensis and Ocimum tenuiflorum, make very little difference, reinforcing the model's very good overall accuracy of 99.07%.

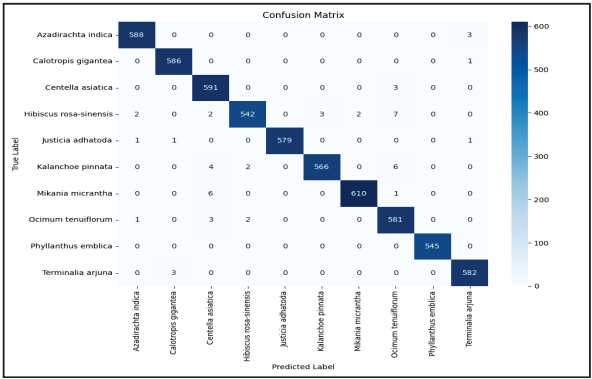


Fig. 5 Confusion matrix for Xception model

F. Comparative Performance Analysis of Transfer Learning Models

Table VI demonstrates that all five models were very good, with just modest metric variances. MobileNetV2 was the leader with 99.43% accuracy and commensurate high precision, recall, and F1-score, reflecting superior consistency. InceptionResNetV2 was a close second (99.34%), and Xception and VGG16 also over 99%, with Xception muddling slightly ahead in the case of precision and recall. DenseNet201 achieved 96.08%, ranking behind the other models. MobileNetV2 demonstrated the most consistent performance, with InceptionResNetV2 and Xception following closely.

TABLE VI. PERFORMANCE COMPARISON OF TRANSFER LEARNING MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
MobileNetV2	99.43	99.43	99.42	99.43
InceptionResNetV2	99.34	99.36	99.34	99.35
Xception	99.07	99.08	99.06	99.07
VGG16	99.00	99.00	99.00	99.00
DenseNet201	96.08	96.21	96.13	96.11

G. Comparison with the existing works

Table VII shows that the proposed models exceed most previous methods in the classification of medicinal plants. Although previous methods like DenseNet201 (96.30%) and InceptionResNetV2 (90.10%) produced moderate accuracy, and specialized models like MedPlantNet (98.51%) and MSFFAR-Net (98.12%) showed improved performance. MobileNetV2 achieved the highest accuracy (99.43%), followed closely by InceptionResNetV2 (99.34%), Xception (99.07%), and VGG16 (99%), with the proposed DenseNet201 (96.08%) matching its earlier performance. These findings confirm the efficiency of fine-tuned transfer learning, especially MobileNetV2 and InceptionResNetV2, in the identification of medicinal plants.

TABLE VII. COMPARITIVE ANALYSIS OF EXPERIMENTED MODELS WITH EXISTING STUDIES

Author	Method	Accuracy	Reference
Islam <i>et al.</i>	DenseNet 201	96.30%	[15]
	InceptionResNetV2	90.10%	
Himel <i>et al.</i>	MedPlantNet	98.51%	[16]
Shanto <i>et al.</i>	CNN-SVM	94.11%	[17]
Islam <i>et al.</i>	MSFFAR-Net	98.12%	[18]

Present Work	DenseNet 201	96.08%
	InceptionResNetV2	99.34%
	VGG16	99%
	MobileNetV2	99.43%
	Xception	99.07%

V. Conclusion

The work shows the potential of transfer learning-based deep learning models in tasks related to the classification of medicinal plant leaves, with state-of-the-art-competitive performance using the vast BDMediLeaves corpus of 38,606 images of ten species. The app-based MobileNetV2-based architecture was the most accurate with an accuracy of 99.43%, superior to other tested architecture—VGG16, DenseNet201, InceptionResNetV2, and Xception, while being as compact as possible for browser and app-based deployment. In addition, the research provides a systematic and reproducible benchmarking platform that lays the foundation for future research in efficient plant recognition systems. By connecting classical botanical expertise and AI-driven innovation, the research facilitates accurate plant identification, sustainable use, and eventual incorporation into digital healthcare and conservational applications.

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